Weekly Progress Report

Week 02 — Data Science & Machine Learning Internship

STUDENT NAME

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DOMAIN

Data Science and Machine Learning

SELECTED PROJECTS

- **Agriculture:** Prediction of Agriculture Crop Production in India
- **Smart City:** Forecasting of Smart City Traffic Patterns

1. Tasks & Progress

- **Milestones Achieved:**
  - Successfully executed Data Cleaning, Missing Value Imputation, and Exploratory Data Analysis (EDA) across both datasets.
  - Structured and prepared datasets for advanced feature engineering and baseline machine learning model training.
- **Key Work Completed:**
  - **Agriculture Project:** Cleaned historical crop yield datasets, addressed missing observations, encoded categorical features (State, Crop Type, Season), and analyzed regional production trends.
  - **Smart City Traffic Project:** Cleaned time-series traffic records, applied noise smoothing, engineered temporal features (hour, day, month), and generated statistical profiles for peak vs. non-peak traffic flow.

2. Challenges & Hurdles

- **Obstacles Encountered:**
  - High multicollinearity observed among consecutive hourly traffic readings in the Smart City dataset.
  - Significant positive skewness in crop production volumes across different states and regional sizes.
- **Solutions Applied:**
  - Implemented Principal Component Analysis (PCA) to reduce feature redundancy while retaining maximum variance.
  - Applied log transformations along with Standardization and Min-Max scaling to achieve balanced feature distributions.

3. Lessons Learned

- **Pandas Operations:** Mastered efficient DataFrame manipulations using `df.drop()`, `df.merge()`, and indexing techniques for large-scale dataset aggregation.
- **Dimensionality Reduction:** Developed a clear understanding of Principal Component Analysis (PCA) and scree plot evaluation for optimal feature selection.
- **Outlier Identification:** Gained hands-on experience using box plots and scatter plots to detect and handle data anomalies in tabular and time-series data.